LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE - 11

MATERI PEMBELAJARAN

Setup Dasar Minikuber & Kubernets

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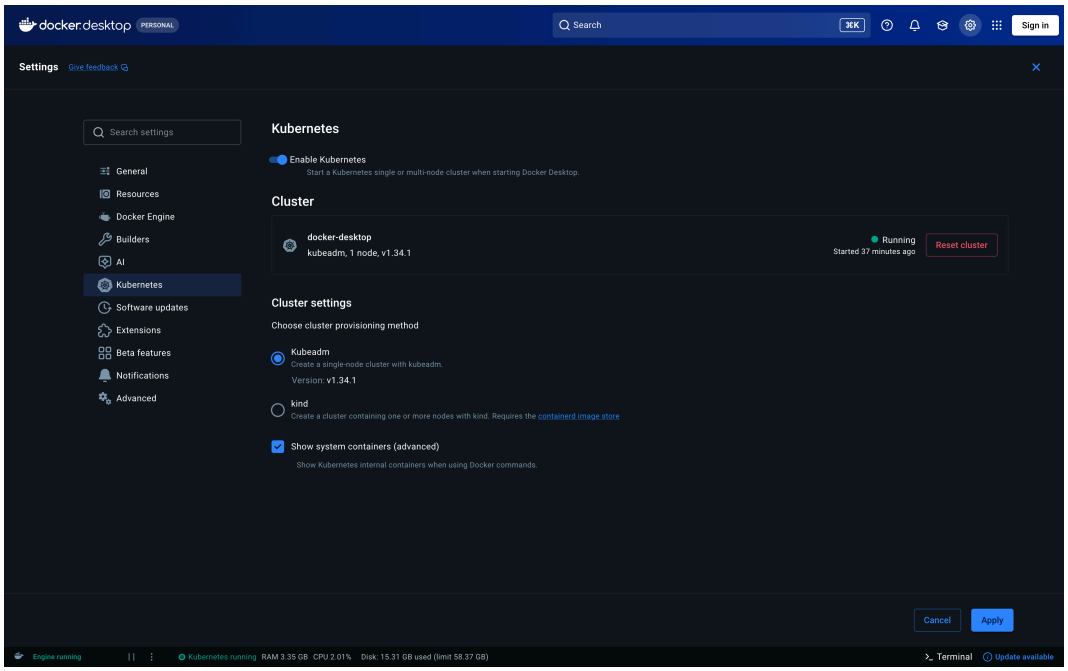
BAB 1 HASIL & PEMBAHASAN

1.1 Instalasi MiniKube

1. Menjalankan Kurbenetes



Gambar 1.1 jalanakan Kubernetes

2. Install Kurbenetes

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % brew install minikube
==> Auto-updating Homebrew...
Adjust how often this is run with `$HOMEBREW_AUTO_UPDATE_SECS` or disable with
`$HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `$HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Downloading https://ghcr.io/v2/homebrew/core/portable-ruby/blobs/sha256:8ad98d9cf15477d1a43f4012cfd5eff5c271398bca1e2724b55bd4944a0bb1d4
################################################################################################################################################ 100.0%
==> Pouring portable-ruby-4.0.3.arm64_big_sur.bottle.tar.gz
==> Auto-updated Homebrew!
==> Updated Homebrew from 5.1.6 (0269fbfafc) to 5.1.8 (424edfbd76).
Updated 2 taps (hashicorp/tap and mongodb/brew).
```

Gambar 1.2 install minikube

3. Jalankan Kurbenetes

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube start --driver=docker
😄  minikube v1.38.1 on Darwin 14.6.1 (arm64)
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🤷  docker "minikube" container is missing, will recreate.
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.3 jalankan kubernetes

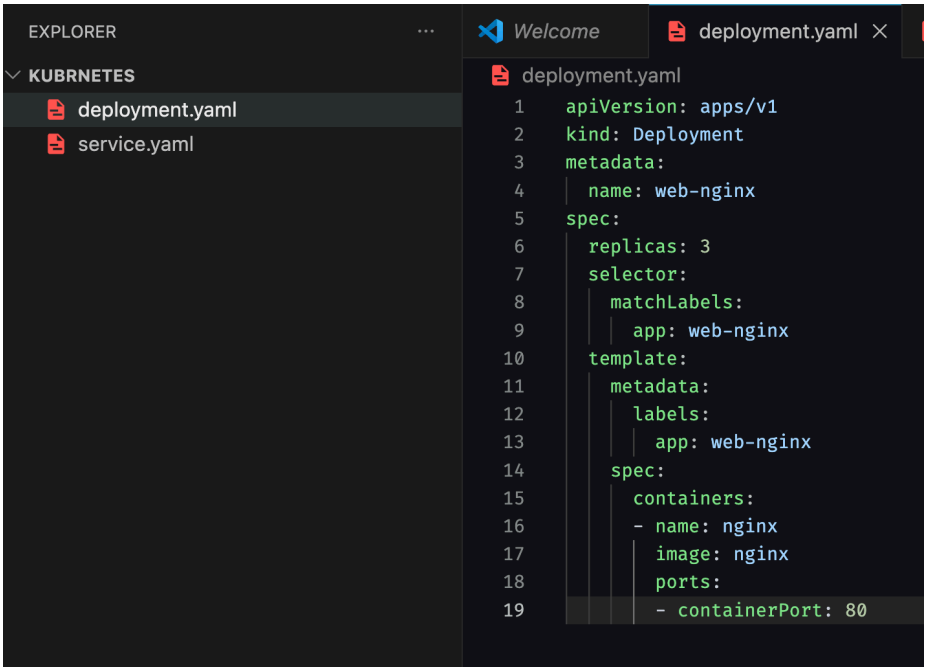
4. Cek Node Kurbenetes

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m13s   v1.35.1
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.4 cek node

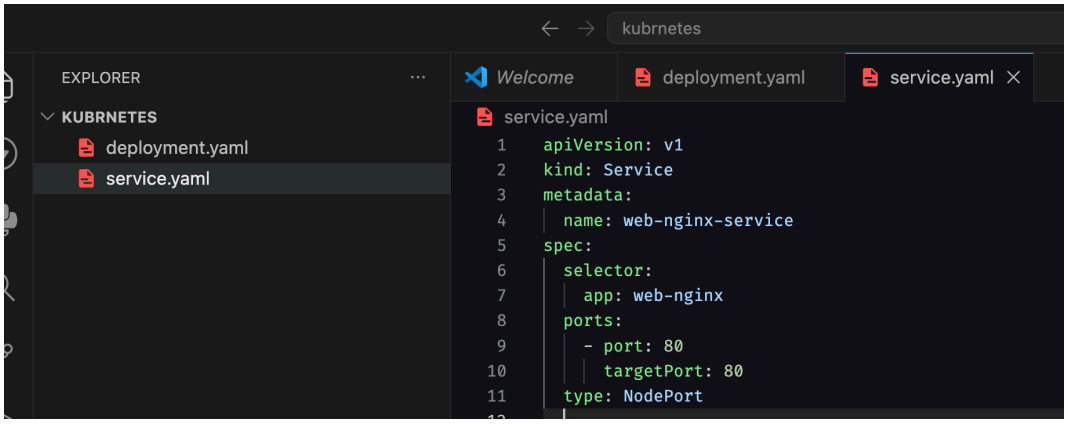
1.2 Deployment & service

1. Membuat deployment.yaml



Gambar 1.4 buat file depolyment.yml

2. Membuat service.yaml



Gambar 1.4 buat file serivce.yml

3. Menjalankan YAML

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl apply -f service.yaml
service/web-nginx-service created
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.5 apply file serivce.yml

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl apply -f deployment.yaml

deployment.apps/web-nginx created
```

Gambar 1.6 apply file deployment.yml

4. Lihat Semua Resource

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/web-nginx-748d764f7-4bsdf     1/1     Running   0          2m1s
pod/web-nginx-748d764f7-954nf     1/1     Running   0          2m1s
pod/web-nginx-748d764f7-g8l6z     1/1     Running   0          2m1s

NAME                          TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1     <none>        443/TCP        6m17s
service/web-nginx-service     NodePort    10.110.80.7   <none>        80:30349/TCP   103s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     3/3     3            3           2m1s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         3       2m1s
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.7 Lihat Semua Resource

5. Cek Pods

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-4bsdf     1/1     Running   0          2m27s
web-nginx-748d764f7-954nf     1/1     Running   0          2m27s
web-nginx-748d764f7-g8l6z     1/1     Running   0          2m27s
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.8 Cek Pods

6. Monitoring Live

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-4bsdf     1/1     Running   0          3m5s
web-nginx-748d764f7-954nf     1/1     Running   0          3m5s
web-nginx-748d764f7-g8l6z     1/1     Running   0          3m5s
```

Gambar 1.9 Monitoring Live

7. Jika status Container Creating terlalu lama:

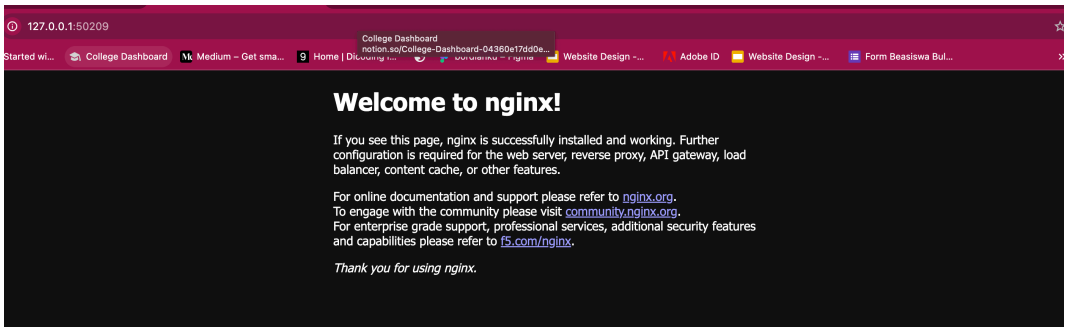
```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl describe pod web-nginx-748d764f7-4bsdf
Name:             web-nginx-748d764f7-4bsdf
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Wed, 29 Apr 2026 10:00:21 +0700
Labels:           app=web-nginx
                  pod-template-hash=748d764f7
Annotations:      <none>
Status:           Running
IP:               10.244.0.6
IPs:
  IP:           10.244.0.6
Controlled By:  ReplicaSet/web-nginx-748d764f7
Containers:
  nginx:
    Container ID:   docker://d971b95952b2a08c1982e39b05a15ea8369b4e7483edc0401eed38b853f29d26
```

Gambar 1.10 lakukan ini jika status Container Creating terlalu lama

8. Membuka Aplikasi

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube service web-nginx-service
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service | 80          | http://192.168.49.2:30349 |
|-----------|-----------------|-------------|---------------------------|
🔗  Starting tunnel for service web-nginx-service.
|-----------|-----------------|-------------|---------------------------|
| NAMESPACE |      NAME       | TARGET PORT |           URL             |
|-----------|-----------------|-------------|---------------------------|
| default   | web-nginx-service |             | http://127.0.0.1:50209    |
|-----------|-----------------|-------------|---------------------------|
🎉  Opening service default/web-nginx-service in default browser...
❗  Because you are using a Docker driver on darwin, the terminal needs to be open to run it.
```

Gambar 1.11 membuka aplikasi



Gambar 1.12 membuka aplikasi

9. Scaling Pod, Lihat pod Live

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-4bsdf   1/1     Running   0          6m9s
web-nginx-748d764f7-954nf   1/1     Running   0          6m9s
web-nginx-748d764f7-g8l6z   1/1     Running   0          6m9s
```

Gambar 1.13 Lihat pod Live

10. Scale Up ke 5

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.14 Scale Up ke 5

11. Hasiil Scale up ke 5 & Lihat Deployment Live

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   5/5     5            5           7m13s
```

Gambar 1.15 membuka aplikasi

1.3 Menambah Worker Baru

1. Tamabh Worker node

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.16 tambah node

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   13m   v1.35.1
minikube-m02   NotReady   <none>          46s   v1.35.1
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.16 get node cek sudah ketambah

```
minikube-m02   NotReady   <none>          46s   v1.35.1
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.17 cek status node baru

1.3 Menhapus & Reset Project

1. Hapus Deployment & Service

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.18 menghapus deployment & service yml

2. Reset Cluster Jika Error

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube delete
🔥  Deleting "minikube" in docker ...
🔥  Deleting container "minikube" ...
🔥  Deleting container "minikube-m02" ...
🔥  Removing /Users/sartriaardianthauno/.minikube/machines/minikube ...
🔥  Removing /Users/sartriaardianthauno/.minikube/machines/minikube-m02 ...
💀  Removed all traces of the "minikube" cluster.
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.19 hapus minikube

```
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes % minikube start --driver=docker
😄  minikube v1.38.1 on Darwin 14.6.1 (arm64)
✨  Using the docker driver based on user configuration
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker Desktop driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
sartriaardianthauno@Satrias-MacBook-Pro kubrnetes %
```

Gambar 1.20 membuat ulang minikube